

Beyond Price Dynamics: Universal Exponents, Phase Transitions, and a New Spectral Framework in Operator Space

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Abstract

We propose that the temporal covariance operator $\mathbf{K}(\tau)$, constructed from the Hankel embedding of logarithmic price returns, provides a natural dynamical representation of financial markets, with the observed price series being a one-dimensional projection of its spectral decomposition. Decomposing $\mathbf{K}(\tau)$ relative to the Marchenko-Pastur (MP) bulk boundary λ_+ , we show that the integrated covariance matrix $\mathbf{X}(\tau) = \int_0^\tau \mathbf{K}(\tau') d\tau'$ exhibits two distinct diffusive regimes: structural modes diffuse with exponent $\beta_s \approx 1.0$, which indicates near Brownian motion, and bulk modes that are anomalously superdiffusive with $\beta_b \in [1.3, 1.7]$ are consistent across two intraday timescales. This spectral separation is explained by a generalized matrix Langevin equation (GLE) with a quasi-block-diagonal memory kernel, the structural block has a finite memory integral (finite effective friction), while the bulk block carries a divergent power-law memory. A third universal exponent $\alpha \approx 2$ governs the power-law distribution of inter-regime waiting times. All three exponents, together with an intrinsic embedding dimension L_{opt} and observation step $\Delta\tau_{\text{opt}}$, are consistent across two intraday timescales (M5 and H1) differing by a factor of 12 in physical time, and across 14 assets spanning equities, indices, foreign exchange, commodities and fixed-income markets. This supports operator-space dynamics as a natural level of description for financial markets and identify the universality class of financial spectral phase transitions.

Keywords: econophysics; random matrix theory; anomalous diffusion; generalized Langevin equation; financial markets; spectral phase transitions; Marchenko-Pastur law.

I Introduction

Financial price series exhibit well-documented stylised facts that resist a unified description, including persistent intra-regime patterns, abrupt structural breaks, volatility clustering, non-stationarity, and anomalous diffusion of returns on multiple timescales [1, 2, 3]. Existing frameworks address these features in isolation: GARCH models capture volatility clustering [4, 5]; Hidden Markov Models describe

regime switching [6]; Random Matrix Theory (RMT) filters noise from *cross-sectional* covariance matrices [7, 8]. None provides a single dynamical equation from which all observed phenomena emerge simultaneously.

The root cause of this fragmentation is a modelling choice, not a fundamental limitation, all these approaches treat the scalar price (or return) series $P(t)$ as the primary dynamical object. Here we take a different stance. We identify the *temporal covariance operator* $\mathbf{K}(\tau) \in \mathbb{R}^{L \times L}$, constructed from the Hankel embedding [9, 10] of a single return series within a sliding window indexed by slow time τ , as an effective dynamical object capturing the evolving correlation structure. The observed return at internal time t within the embedding window centred at slow time τ admits the spectral reconstruction

$$P(t; \tau) = \underbrace{\sum_{k: \lambda_k(\tau) > \lambda_+(\tau)} a_k(\tau) \phi_k(t; \tau)}_{\text{structural signal}} + \underbrace{\sum_{k: \lambda_k(\tau) \leq \lambda_+(\tau)} a_k(\tau) \phi_k(t; \tau)}_{\text{bulk noise}}, \quad (1)$$

where $t \in \{1, \dots, L\}$ is the internal (fast) lag index within the embedding window—distinct from the slow time τ that indexes successive windows— $\{\lambda_k(\tau), \phi_k(t; \tau)\}$ are the eigenpairs of $\mathbf{K}(\tau)$, $\lambda_+(\tau)$ is the Marchenko–Pastur upper boundary, and

$$a_k(\tau) = \sum_{t=1}^L P(t; \tau) \phi_k(t; \tau) \quad (2)$$

are the projection scores of the return series onto the k -th eigenmode within window τ . This decomposition is directly analogous to the distinction between a density matrix, the fundamental object of quantum mechanics, and the scalar expectation values accessible to any particular measurement.

The apparent complexity of $P(t)$, including long memory, non-stationarity, and volatility clustering, can be interpreted as arising from projecting a high-dimensional operator process onto a one-dimensional observable. Our main result is that $\mathbf{K}(\tau)$ obeys a *generalized matrix Langevin equation* (GLE) with a quasi-block-diagonal memory kernel, predicting two distinct diffusion exponents for the structural and bulk subspaces, together with a universal first-passage exponent for spectral phase transitions. Empirical analysis across 14 assets and five asset classes confirms three universal exponents: $(\beta_s, \beta_b, \alpha)$.

II Construction of the Covariance Operator

Let $P(t) = \ln[S(t)/S(t-1)] - \mu(\tau)$ be the mean-centred log-return, where $\mu(\tau)$ is the sample mean within a window of length W centred at slow time τ . Fix an embedding dimension $L < W$ and define the Hankel matrix

$$\mathbf{H}(\tau) \in \mathbb{R}^{N \times L}, \quad H_{ki} = P(\tau - W + k + i - 1), \quad N = W - L. \quad (3)$$

The empirical covariance operator is

$$\mathbf{K}(\tau) = \frac{1}{N-1} [\mathbf{H}(\tau)^\top \mathbf{H}(\tau) - N \bar{h} \bar{h}^\top] \in \mathbb{R}^{L \times L}, \quad (4)$$

with $\bar{h}$ the column-mean vector. Eigendecomposition yields $\mathbf{K}(\tau) \varphi_k(\tau) = \lambda_k(\tau) \varphi_k(\tau)$, ordered $\lambda_1 \geq \dots \geq \lambda_L$. The MP upper bulk boundary is

$$\lambda_+(\tau) = \hat{\sigma}^2(\tau) (1 + \sqrt{q})^2, \quad q = L/N, \quad (5)$$

with $\hat{\sigma}^2(\tau)$ the empirical variance of $\mathbf{H}(\tau)$. Modes with $\lambda_k > \lambda_+$ carry structural information, the remainder constitute the bulk. The instantaneous structural dimension is $m(\tau) = \#\{k : \lambda_k(\tau) > \lambda_+(\tau)\}$.

The structural and bulk projectors are $\mathbf{P}_s(\tau) = \sum_{k:\lambda_k > \lambda_+} \varphi_k \varphi_k^\top$ and $\mathbf{P}_b(\tau) = \mathbf{I} - \mathbf{P}_s(\tau)$. We define the structural and bulk components of the integrated process

$$\mathbf{X}(\tau) = \sum_{\tau' < \tau} \mathbf{K}(\tau') \Delta\tau \quad (6)$$

, where $\Delta\tau$ is the time between observations, and $\mathbf{X}_s = \mathbf{X}\mathbf{P}_s$, $\mathbf{X}_b = \mathbf{X}\mathbf{P}_b$. The mean-square displacement (MSD) of each component,

$$\text{MSD}_{s,b}(s) = \mathbb{E} \left[\|\mathbf{X}_{s,b}(\tau + s) - \mathbf{X}_{s,b}(\tau)\|_F^2 \right] \sim s^{\beta_{s,b}}, \quad (7)$$

, where F stands for Frobenius norm, defines the diffusion exponents in the asymptotic regime.

III Generalized Matrix Langevin Equation

A Equation of motion

We model the evolution of $\mathbf{K}(\tau)$ on the slow timescale by the matrix GLE

$$\frac{\Delta \mathbf{K}}{\Delta\tau} = - \sum_{\tau' < \tau} \mathbf{\Gamma}(\tau - \tau') \mathbf{K}(\tau') \Delta\tau + \boldsymbol{\xi}(\tau) + \delta \mathbf{K} \Delta N_\tau, \quad (8)$$

where $\mathbf{\Gamma}(\tau)$ is the matrix memory kernel, $\boldsymbol{\xi}(\tau)$ is coloured noise related to the kernel by the fluctuation-dissipation theorem (FDT) [11, 12, 13]

$$\mathbb{E}[\boldsymbol{\xi}(\tau) \boldsymbol{\xi}^\top(0)] = \|\mathbf{K}\|_F^2 \mathbf{\Gamma}(\tau), \quad (9)$$

where $\|\mathbf{K}\|_F^2 = \text{Tr}(\mathbf{K}^\top \mathbf{K})$ is the squared Frobenius norm of $\mathbf{K}$, playing the role of an effective temperature that sets the fluctuation amplitude proportional to the current spectral energy of the operator, and ΔN_τ is a Poisson process for large exogenous shocks. The noise is coloured, not white Gaussian, it carries the same temporal structure as the kernel, and the two are linked by the FDT as a single object.

The GLE in Eq. (8) is not introduced as a phenomenological ansatz but follows from a Mori–Zwanzig projection onto the operator subspace, see Supplementary Material (SM) Section S1 for the explicit construction. As demonstrated in SM, the memory kernel $\mathbf{\Gamma}(\tau)$ and the random force $\boldsymbol{\xi}(\tau)$ are coupled via a matrix-valued Fluctuation-Dissipation Theorem (FDT, Eq(9)), ensuring consistency with causality and energy balance in the Frobenius metric.

B Quasi-block-diagonal memory kernel

To account for the coexistence of $\beta_s \neq \beta_b$, the kernel must carry different memory structures in the two spectral subspaces. In the instantaneous eigenbasis of $\mathbf{K}$ we propose that

$$\mathbf{\Gamma}(\tau) \approx \begin{pmatrix} \mathbf{\Gamma}_s(\tau) & \boldsymbol{\varepsilon}(\tau) \\ \boldsymbol{\varepsilon}^\top(\tau) & \mathbf{\Gamma}_b(\tau) \end{pmatrix}, \quad \|\boldsymbol{\varepsilon}\| \ll \|\mathbf{\Gamma}_s\|, \|\mathbf{\Gamma}_b\|. \quad (10)$$

The weak off-diagonal block $\boldsymbol{\varepsilon}$ couples the two subspaces and induces a small observed excess $\beta_s - 1 > 0$.

For the *structural block*, the key condition is a convergent memory integral [14]:

$$\gamma_s^{\text{eff}} = \sum_{n=0}^{\infty} \|\mathbf{\Gamma}_s(n \Delta\tau)\|_F \Delta\tau < \infty. \quad (11)$$

A finite, nonzero Frobenius-norm integral is both necessary and sufficient for normal diffusion ($\beta_s = 1$) in the asymptotic regime [14], *regardless* of whether the memory itself is short or long-ranged. The detailed shape of $\Gamma_s(\tau)$ at intermediate times does not matter; only its integral does. In the asymptotic limit, Γ_s acts as an effective delta-function friction $\gamma_s^{\text{eff}} \delta(\tau)$.

For the *bulk block*, we assume a power-law memory:

$$\|\Gamma_b(\tau)\|_F \sim \tau^{-(2-\nu)}, \quad 0 < \nu < 1, \quad (12)$$

whose Frobenius-norm integral diverges. A divergent integral with this power-law form produces anomalous superdiffusion $\beta_b = 1 + \nu$ [14]. Physically, the $L \gg m(\tau)$ bulk modes are quasi-degenerate and strongly coupled, their collective stochastic mixing via Dyson-type eigenvalue dynamics [15, 16] generates effective long-range correlations in the bulk sector even though each individual mode is locally Markovian.

The diffusion exponents β_s and β_b characterize the scaling of the integrated operator $X(\tau) = \int_0^\tau \mathbf{K}(s) ds$. By projecting the matrix dynamics onto the Frobenius norm $\|\cdot\|_F$, we establish (see SM, Section S2) that β is uniquely determined by the integrability of the memory kernel $\Gamma(\tau)$.

C Asymptotic diffusion exponent

Following [14], $\beta_{s,b}$ is estimated by log-log regression of $\text{MSD}_{s,b}(s)$ restricted to the upper 70% of available lags. Discarding the lower 30% where transient behaviour dominates.

The small excess $\beta_s > 1$ (Table 1) is attributed to three sources in decreasing order of expected magnitude: (i) weak inter-block coupling ε , which injects a superdiffusive component from the bulk; (ii) the Poisson term $\delta \mathbf{K} \Delta N_\tau$, contributing sporadic large increments; and (iii) finite-sample regression uncertainty.

D Fluctuation–dissipation balance

Applying Itô’s lemma to the Frobenius energy $E = \|\mathbf{K}\|_F^2 = \text{Tr}(\mathbf{K}^\top \mathbf{K})$, the increment decomposes as

$$\Delta E = \underbrace{2 \text{Tr}(\mathbf{K} \Delta \mathbf{K}^\top)}_{\text{linear term}} + \underbrace{\text{Tr}(\Delta \mathbf{K} \Delta \mathbf{K}^\top)}_{\text{quadratic term}}, \quad (13)$$

with no approximation. The linear term is odd in $\Delta \mathbf{K}$ and changes sign with the direction of motion of $\mathbf{K}$; near stationarity it acts as an energy sink. The quadratic term is positive definite and acts as a source. We therefore define

$$D(\tau) = -2 \text{Tr}(\mathbf{K}(\tau) \Delta \mathbf{K}(\tau)^\top), \quad (14)$$

$$I(\tau) = \text{Tr}(\Delta \mathbf{K}(\tau) \Delta \mathbf{K}(\tau)^\top), \quad (15)$$

as the instantaneous dissipation and injection rates, and the energy-balance ratio as

$$R_{\text{FDT}} = \frac{\langle D \rangle}{\langle I \rangle}, \quad (16)$$

where $\langle \cdot \rangle$ denotes a time average. At stationarity, $\langle \Delta E \rangle = 0$ implies $\langle D \rangle = \langle I \rangle$ and hence $R_{\text{FDT}} = 1$ exactly, by Eq. (13). This result follows solely from the Itô decomposition and holds independently of the diffusive regime: in particular, it does not require the effective friction to be finite, and remains valid in the superdiffusive regime where the effective friction vanishes. Deviations of R_{FDT} from unity therefore signal departures from stationarity rather than transitions between dynamical regimes. Empirically, all liquid assets yield $R_{\text{FDT}} \approx 1.0$ to within the measurement uncertainty (Table 1), confirming that the operator $K(\tau)$ operates near a stationary state over the observation window, consistently with the stationarity assumption underlying the Hankel embedding.

IV Spectral Phase Transitions

A Regime changes as first-passage events

Define the spectral gap $G(\tau) = \lambda_{m(\tau)}(\tau) - \lambda_+(\tau)$: the signed distance between the marginal structural eigenvalue and the Marchenko–Pastur boundary. A regime change (increment or decrement of $m(\tau)$) occurs when $G(\tau) = 0$, so the sequence of inter-regime durations $\{\tau_k^*\}$ is precisely the sequence of first-passage times of G through zero. Applying the GLE eigenvalue calculus [15, 16], the gap evolves as

$$\Delta G = \mu(G, \tau) \Delta \tau_{\text{opt}} + \sigma(G, \tau) \Delta W_\tau, \quad (17)$$

with $\mu(G) = a(\tau)G + \mathcal{O}(G^2)$ and $\sigma(0, \tau) > 0$. As $G \rightarrow 0$, the drift is linear while the diffusion coefficient remains finite, which places the transition in the universality class of one-dimensional diffusions at a regular critical point [17, 18].

For such processes, first-passage theory predicts a power-law distribution of return times $g(\tau^*) \sim (\tau^*)^{-\alpha}$ with the exponent controlled by the local drift near the boundary [17, 18]. In the marginal regime where the drift vanishes, values near $\alpha \approx 2$ are expected. We therefore treat α as an empirical exponent, and verify that the measured values $\hat{\alpha} \in [1.87, 2.45]$ are consistent with this scenario.

B Adiabatic eigenvector dynamics

When $\mathbf{K}(\tau)$ varies slowly in τ , differentiating the eigenvalue equation $\mathbf{K}(\tau)\phi_k(\cdot; \tau) = \lambda_k(\tau)\phi_k(\cdot; \tau)$ with respect to τ and projecting onto ϕ_j ($j \neq k$) gives

$$\partial_\tau \phi_k = \sum_{j \neq k} \frac{\phi_j^\top (\partial_\tau \mathbf{K}) \phi_k}{\lambda_k - \lambda_j} \phi_j, \quad (18)$$

the discrete analogue of adiabatic perturbation theory for self-adjoint operators [19, ?]. Here the diagonal term $j = k$ vanishes by differentiation of the normalisation condition $\|\phi_k\|^2 = 1$. Equation (18) applies to structural modes ($\lambda_k > \lambda_+$); eigenvectors within the MP bulk are stochastically mixed and do not admit a smooth adiabatic trajectory. As $\lambda_k \rightarrow \lambda_j$, the denominator vanishes and the eigenvectors rotate rapidly—the spectral precursor of a regime change. The order parameter of the transition is

$$\Delta(\tau) = \min_{k \neq j} |\lambda_k(\tau) - \lambda_j(\tau)|. \quad (19)$$

V Bulk Spectral Universality

In the quasi-degenerate bulk subspace, stochastic mixing via (18) induces Brownian rotation of eigenvectors. Projections of returns onto bulk modes become approximately Gaussian, and the spectral density converges to the MP law [20, 21]:

$$\rho_{\text{MP}}(\lambda) = \frac{\sqrt{(\lambda_+ - \lambda)(\lambda - \lambda_-)}}{2\pi q \hat{\sigma}^2 \lambda}, \quad (20)$$

where $\lambda_\pm = \hat{\sigma}^2(1 \pm \sqrt{q})^2$.

The observed relative L^2 error of $\approx 3.7\%$ between the empirical bulk histogram and (20) is attributed to short-range spurious correlations introduced by the overlapping Hankel window construction, which slightly deforms the bulk spectrum. This systematic offset is a finite-size construction artefact and does not affect the structural eigenvalue dynamics.

VI Empirical Results

We analyse log-returns for 14 assets across equities, fixed-income, foreign exchange, commodity and rates markets (Table 1) at two intraday timescales: M5 (5-minute bars) and H1 (1-hour bars), each spanning 5 years. We choose the optimal parameters according to a robustness criterion defined in subsection (E).

A Bulk spectral density (Fig. 1)

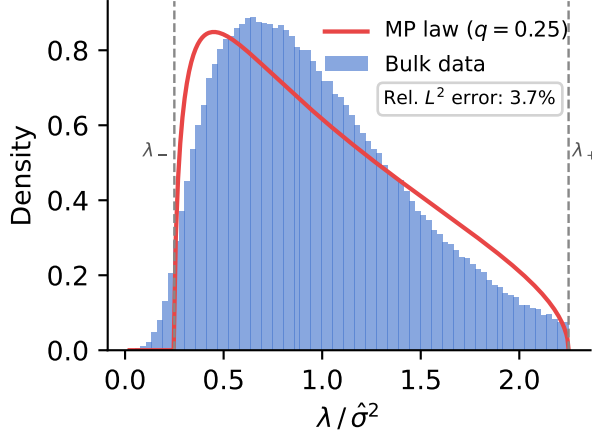


Figure 1: **Bulk eigenvalue density, PETR4 (M5).** Aggregated, normalised bulk eigenvalue histogram (blue bars). Red curve: theoretical MP density Eq. (20) with $q = 0.25$, $\hat{\sigma}^2 = 1$. Vertical dashed lines: $\lambda_{\pm}$. *Inset:* relative L^2 error between empirical bulk histogram and MP law. The systematic $\approx 3.7\%$ deviation is attributed to short-range spurious correlations from the overlapping Hankel window construction.

Figure 1 shows the aggregated, normalised bulk eigenvalue histogram for PETR4 (M5 timescale). The relative L^2 error to the theoretical MP density (20) is 3.7% (inset). The small systematic deviation is attributed to Hankel window overlap.

B Structural diffusion exponent β_s (Fig. 2a,c)

Figure 2(a) shows the log-log MSD of $\mathbf{X}_s(\tau)$ for PETR4 at M5 and H1. Both timescales yield $\hat{\beta}_s \approx 1.0$, nearly a Brownian motion. The exponent is estimated over the upper 70% of lags [14]. The small excess $\delta\beta_s = \hat{\beta}_s - 1 > 0$ is consistent with weak bulk-structure coupling and the Poisson term (Sec. III.C).

Figure 2(c) shows $\hat{\beta}_s$ vs. observation step $\Delta\tau_0$. For $\Delta\tau_0 < \Delta\tau_{\text{opt}}$, overlapping successive windows bias the MSD upward; for $\Delta\tau_0 > \Delta\tau_{\text{opt}}$, M5 and H1 decouple into their respective physical asymptotic regimes, breaking inter-scale coherence. The optimal range $\Delta\tau_{\text{opt}} \approx 15\text{-}25$ candles is the regime where successive increments are approximately uncorrelated and both timescales share the same universal dynamics.

C Bulk diffusion exponent $\hat{\beta}_b$

The bulk MSD exponent $\hat{\beta}_b$ is superdiffusive for all assets and both timescales (Table 1), with values in the range $[1.3, 1.7]$. This spread reflects genuine differences in the collective eigenvalue dynamics of the

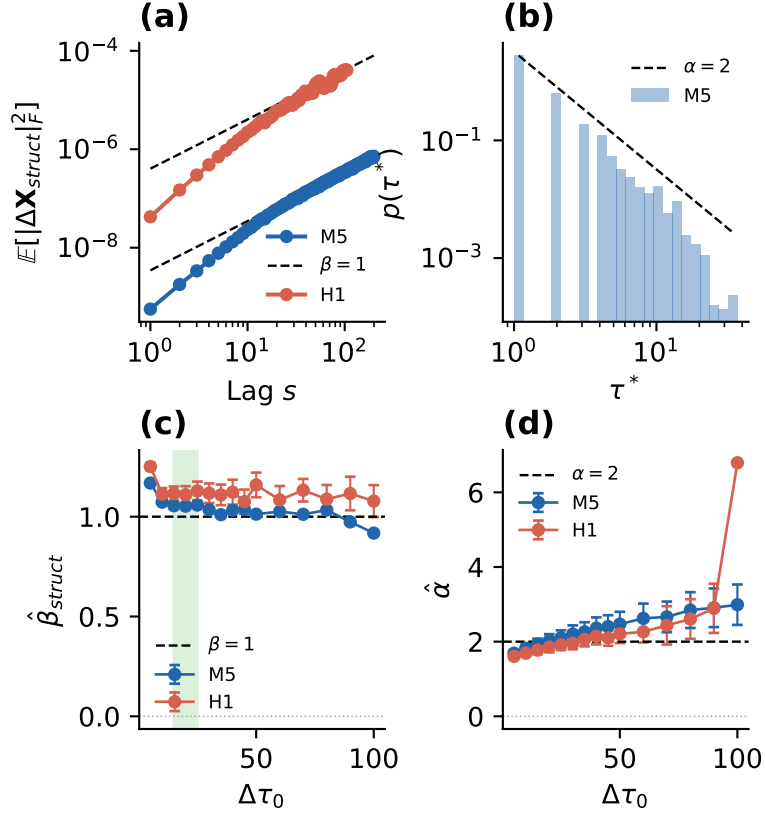


Figure 2: **Universal exponents, PETR4.** (a) Log-log MSD of $\mathbf{X}_s(\tau)$ vs. lag s (in units of $\Delta\tau_{opt}$) at M5 (blue) and H1 (red). Dashed: reference slope $\beta = 1$. Asymptotic exponent $\hat{\beta}_s \approx 1.0$ estimated over the upper 70% of lags [14]. (b) Log-log histogram of inter-regime durations τ^* at M5; dashed: slope $\alpha = 2$. (c) $\hat{\beta}_s$ vs. observation step $\Delta\tau_0$; green band: optimal range $\Delta\tau_{opt} \approx 15-25$ candles. (d) $\hat{\alpha}$ vs. $\Delta\tau_0$; dashed: $\alpha = 2$.

bulk sector across asset classes and sampling frequencies: assets with denser microstructure or higher liquidity tend toward higher values of $\hat{\beta}_b$, consistent with stronger Dyson-type stochastic mixing among quasi-degenerate bulk modes (Sec. B). The central tendency $\hat{\beta}_b \approx 1.5$, is representative of the liquid equity and FX futures sample; the AR(1) reference processes yield $\hat{\beta}_b \approx \hat{\beta}_s \approx 1$ for all φ (Supplementary Table 1), confirming that the structural–bulk exponent separation is a genuine dynamical signature of real markets and not an artefact of the Hankel embedding.

D Inter-regime waiting times α (Fig. 2b,d)

Figure 2(b) shows the log-log histogram of inter-regime intervals τ^* for PETR4 at M5, where a power-law tail with $\hat{\alpha} \approx 2$ is observed. Figure 2(d) shows $\hat{\alpha}$ vs. $\Delta\tau_0$; convergence to ≈ 2 is robust for both timescales at smaller $\Delta\tau_0$. Exponential and Gaussian waiting-time distributions, characteristic of Poisson and GARCH models, are strongly rejected.

E Intrinsic embedding dimension L_{opt} (Fig. 3)

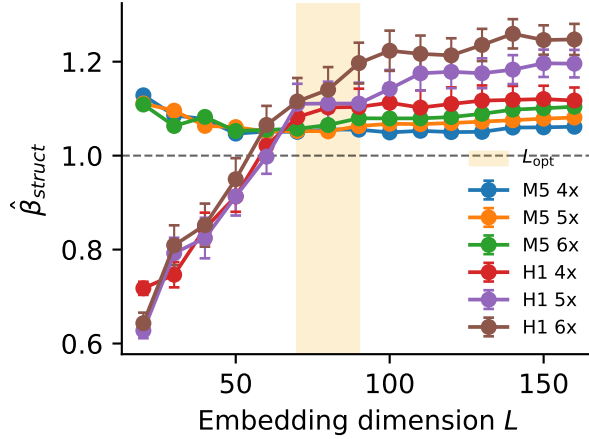


Figure 3: **Intrinsic embedding dimension, PETR4.** $\hat{\beta}_s$ vs. L for $n = W/L \in \{4, 5, 6\}$ at M5 (cool colours) and H1 (warm colours). Yellow band: $L_{\text{opt}} \in [70, 90]$, defined by the inter-scale coherence criterion (28). Both timescales converge in the shaded region, identifying $L_{\text{opt}} \approx 80$ as an intrinsic scale invariant under a factor-of-12 change in physical time.

Figure 3 shows $\hat{\beta}_s$ vs. L for $n = W/L \in \{4, 5, 6\}$ and both timescales. For $L < L_{\text{opt}}$, the spectral projector $\mathbf{P}_s$ is geometrically unstable, the spectral gap is insufficient to isolate structural eigenmodes, producing a subdiffusive artefact in H1 timescale $\hat{\beta}_s < 1$. For $L > L_{\text{opt}}$, lags beyond the process correlation time τ_c contaminate $\mathbf{P}_s$ with bulk eigenmodes.

The parameters $(\Delta\tau_{\text{opt}}, L_{\text{opt}})$ are selected by a cross-scale robustness criterion that requires no theoretical anchor. Both optimizations share the same composite score structure: a local-stability term and an inter-scale coherence term, each normalized by its grid average to render them dimensionless and commensurable without any free parameter.

For $\Delta\tau_{\text{opt}}$, we compute $\hat{\beta}_{\text{struct}}$ at both M5 and H1 for each candidate step and define the composite

score at each interior grid point (excluding the two outermost points on each side) as

$$\mathcal{S}(\Delta\tau_0) = \frac{\bar{\sigma}_{\text{local}}^2(\Delta\tau_0)}{\bar{\sigma}_{\text{local}}^2} + \frac{B(\Delta\tau_0)}{\bar{B}}, \quad (21)$$

where

$$\bar{\sigma}_{\text{local}}^2(\Delta\tau_0) = \frac{1}{2} \sum_{f \in \{\text{M5}, \text{H1}\}} \text{Var} \left[\hat{\beta}_{\text{struct}}^{(f)}(\Delta\tau_0 + jh), j \in \{-2, -1, 0, 1, 2\} \right] \quad (22)$$

is the mean of the local variances computed independently for each timescale over the five-point window centered at $\Delta\tau_0$ (grid step h), and

$$B(\Delta\tau_0) = \left| \hat{\beta}_{\text{struct}}^{(\text{M5})}(\Delta\tau_0) - \hat{\beta}_{\text{struct}}^{(\text{H1})}(\Delta\tau_0) \right| \quad (23)$$

is the inter-scale discrepancy. The grid averages $\bar{\sigma}_{\text{local}}^2$ and $\bar{B}$ are computed over all interior points. The optimal lag is

$$\Delta\tau_{\text{opt}} = \arg \min_{\Delta\tau_0} \mathcal{S}(\Delta\tau_0). \quad (24)$$

The first term selects the plateau of each curve independently, penalizing regions where $\hat{\beta}_{\text{struct}}$ is still varying, while the second penalizes systematic inter-scale discrepancy. Separating the two contributions prevents pathological cases where the curves cross at small $\Delta\tau_0$ (driving $B \rightarrow 0$ artificially) or where they are parallel and offset (making B uninformative). The five-point window (second neighbors) provides robustness against isolated noise spikes that would corrupt a three-point estimate.

For L_{opt} , an analogous composite score is defined, with the sole difference that only first neighbors are used in the local-stability term, reflecting the coarser resolution of the embedding-dimension grid:

$$\mathcal{S}_L(L) = \frac{\bar{\sigma}_{L,\text{local}}^2(L)}{\bar{\sigma}_{L,\text{local}}^2} + \frac{B_L(L)}{\bar{B}_L}, \quad (25)$$

where

$$\bar{\sigma}_{L,\text{local}}^2(L) = \frac{1}{2} \sum_{f \in \{\text{M5}, \text{H1}\}} \text{Var} \left[\hat{\beta}_{\text{struct}}^{(f)}(L + jh_L), j \in \{-1, 0, 1\} \right] \quad (26)$$

is the mean of the local variances over the three-point window (grid step h_L), and

$$B_L(L) = \left| \hat{\beta}_{\text{struct}}^{(\text{M5})}(L) - \hat{\beta}_{\text{struct}}^{(\text{H1})}(L) \right| \quad (27)$$

is the inter-scale discrepancy at fixed multiplier. The optimal embedding dimension is

$$L_{\text{opt}} = \arg \min_L \mathcal{S}_L(L). \quad (28)$$

At L_{opt} , the score is by construction at its minimum with respect to both local stability and inter-scale coherence, the structural exponent is by construction insensitive to the choice of window multiplier within the range $\{4L, 5L, 6L\}$ at that point, and we adopt the window of $5L$ candles as the reference throughout. The two optimizations are performed in series: $\Delta\tau_{\text{opt}}$ is determined first with $L = 70$ fixed, and L_{opt} is then determined with $\Delta\tau_{\text{opt}}$ fixed.

Table 1: **Universal exponents across asset classes and markets.** $\hat{\beta}_s$: structural MSD exponent; $\hat{\alpha}$: inter-regime waiting-time exponent; $\hat{\beta}_b$: bulk MSD exponent; L_{opt} : optimal embedding dimension (timescale-invariant, single column). M5: 5-min bars; H1: 1-hour bars. Asset classes: ■ Equities, ■ Index, ■ Forex, ■ Commodities and ■ Rates.

	M5 (5-min)			H1 (1-hour)			
Asset/Market	$\hat{\alpha}$	$\hat{\beta}_s$	$\hat{\beta}_b$	$\hat{\alpha}$	$\hat{\beta}_s$	$\hat{\beta}_b$	L_{opt}
PETR4/B3	2.023 ± 0.181	1.052 ± 0.006	1.268 ± 0.005	1.849 ± 0.122	1.111 ± 0.046	1.556 ± 0.023	80
ITUB4/B3	2.013 ± 0.170	1.020 ± 0.012	1.603 ± 0.008	1.973 ± 0.153	0.925 ± 0.019	1.546 ± 0.018	100
VALE3/B3	2.247 ± 0.247	0.965 ± 0.010	1.653 ± 0.014	2.245 ± 0.281	0.857 ± 0.063	1.503 ± 0.044	120
AAPL/NYSE	TBD	TBD	TBD	TBD	TBD	TBD	TBD
S&P 500/NYSE	TBD	TBD	TBD	TBD	TBD	TBD	TBD
Nikkei 225/TSE	TBD	TBD	TBD	TBD	TBD	TBD	TBD
CAC 40/XPAR	TBD	TBD	TBD	TBD	TBD	TBD	TBD
WIN\$N/B3	2.254 ± 0.262	1.015 ± 0.007	1.640 ± 0.012	2.312 ± 0.268	1.019 ± 0.041	1.576 ± 0.030	70
EUR/USD/OTC	TBD	TBD	TBD	TBD	TBD	TBD	TBD
WDO\$N/B3	2.240 ± 0.225	1.031 ± 0.005	1.681 ± 0.014	2.184 ± 0.234	0.719 ± 0.066	1.377 ± 0.060	80
Gold/OTC	TBD	TBD	TBD	TBD	TBD	TBD	TBD
WTI/CME	TBD	TBD	TBD	TBD	TBD	TBD	TBD
DI1/B3	2.116 ± 0.197	1.051 ± 0.011	1.370 ± 0.021	1.982 ± 0.160	1.059 ± 0.030	1.333 ± 0.050	30
US 10Y/CME	TBD	TBD	TBD	TBD	TBD	TBD	TBD
Mean	TBD	TBD	TBD	TBD	TBD	TBD	TBD
Std	TBD	TBD	TBD	TBD	TBD	TBD	TBD

F Universal exponents across asset classes (Table 1)

Table 1 summarises $\hat{\beta}_s$, $\hat{\alpha}$, $\hat{\beta}_b$ and L_{opt} for all 14 assets at both timescales. The universality of all three exponents across asset classes, market microstructures and sampling frequencies constitutes strong evidence for a universal dynamical class in operator space.

A notable exception is DI1 (Brazilian overnight interest-rate futures), whose optimal embedding dimension $L_{\text{opt}} = 30$ is substantially smaller than for equities and FX futures ($L_{\text{opt}} \in [70, 120]$, Table 1). A smaller L_{opt} indicates that the dynamical complexity of $\mathbf{K}(\tau)$ is fully resolved by a lower-dimensional state space: structural modes are fewer but more dominant relative to the bulk, which has fewer effective degrees of freedom. This is consistent with the simpler autocorrelation structure of interest-rate futures, whose price dynamics are tightly anchored to monetary policy expectations and exhibit lower intraday complexity than equity or currency markets.

VII Discussion

A The fundamental object and its projection onto prices

Classical approaches to financial dynamics, whether rooted in stochastic calculus [22, 23], econometric volatility models [4, 5], or the statistical physics of price fluctuations [3, 1], take the price process $P(\tau)$, or equivalently the return series $r(\tau)$, as the primary observable. Structure is then inferred by measuring properties of this scalar time series: its distribution, autocorrelation function, variance clustering, and spectral density. The present work proposes a different starting point: the *temporal covariance operator* $K(\tau)$ is the fundamental dynamical object, and the observed price return is its spectral projection onto a one-dimensional observable via Eq. (??).

This identification has immediate interpretive consequences. GARCH-type volatility clustering is the scalar footprint of superdiffusive bulk operator dynamics; regime changes are first-passage events of $G(\tau)$ through zero; and the apparent long memory in scalar returns [24], which has long resisted unambiguous interpretation via Hurst exponents or DFA, is the projection of the superdiffusive bulk onto a one-dimensional observable, not an intrinsic property of prices. Standard scalar tests applied directly to $P(t)$ cannot distinguish these operator-level phenomena. This establishes a clear hierarchy: $K(\tau)$ acts as a generator, while $P(t)$ corresponds to a projected observable.

This shift is not merely a change of representation: it promotes a derived statistical quantity to the role of a dynamical field, in the same spirit that statistical mechanics replaces the trajectory of individual particles with a density field or an order parameter. The operator $K(\tau)$ encodes simultaneously the instantaneous correlation structure of the market, its spectral decomposition into signal and noise, and its temporal evolution, none of which are directly accessible from the scalar price series alone.

The specific form of the generalised matrix Langevin equation derived from this perspective, with a quasi-block-diagonal memory kernel coupling the structural and bulk subspaces of $K(\tau)$, is a new result that emerges directly from the empirical separation of two distinct diffusive regimes.

B Why bulk diffusion is more anomalous than the signal

The result $\hat{\beta}_b \approx 1.3 - 1.7 > \hat{\beta}_s \approx 1.0$ may appear counterintuitive. The resolution lies in the collective nature of bulk diffusion. The $L \gg m$ quasi-degenerate bulk modes are strongly coupled through Eq. (18): near-degeneracy amplifies their mutual rotations. This Dyson-type stochastic mixing [15] generates effective long-range memory in the bulk sector, a divergent $\hat{\Gamma}_b(0)$, even though each individual mode is locally Markovian. By contrast, the few m structural modes are spectrally isolated, weakly coupled, and governed by a memory kernel with a finite integral, yielding near-Brownian diffusion.

C Connections to established frameworks

Random matrix theory in finance. The application of RMT to financial correlation matrices, pioneered by Laloux et al. [7] and Plerou et al. [8], established that the bulk of the empirical spectrum is well described by the Marchenko-Pastur law [20], with a small number of eigenvalues carrying genuine market information. This programme, developed extensively in the context of portfolio optimisation and eigenvalue cleaning [2, 25, 26], treats the spectral decomposition as a static property of the market. The present framework extends this programme in two directions. First, the MP separation is applied locally at each τ , transforming a single static snapshot into a dynamical observable, whose stability is characterised by α , a new dynamical quantity with no counterpart in the static RMT literature. Second, the quasi-block-diagonal memory kernel is the equation of motion that the structural/bulk separation implies, an object the static RMT programme, by construction, could not have produced. These are extensions, not contradictions: the static MP decomposition is recovered as the long-time limit of the local decomposition, and results on eigenvalue cleaning [25] remain fully valid within our framework.

Anomalous diffusion. Building on the anomalous diffusion framework developed in Refs. [14, 27], by analogy with the scalar GLE, where the diffusion exponent β is carried by the *position* $x(t) = \int_0^t v(t') dt'$ rather than by the velocity $v(t)$ itself, the exponent $\beta_{s,b}$ here belongs to the integrated operator process $\mathbf{X}(\tau) = \int_0^\tau \mathbf{K}(\tau') d\tau'$ rather than to $\mathbf{K}(\tau)$ directly. This analogy is formal— $\mathbf{K}(\tau)$ is a positive-semidefinite covariance operator, not a signed velocity field—but it is operationally precise: it identifies $\mathbf{X}(\tau)$ as the correct process from which $\beta_{s,b}$ must be estimated via the log-log slope of $\text{MSD}_{s,b}(s)$, and rules out estimating β directly from the increments of $\mathbf{K}$. This identification, grounded in the fluctuation-dissipation relations [11], is what makes a direct comparison between $\hat{\beta}_s$ and $\hat{\beta}_b$ physically meaningful. A central result of Ref. [14] is that unbiased estimation of β requires computing the MSD of the position,

not of the velocity; the estimation procedure adopted here implements this principle directly, extracting β from the log-log slope of the MSD of the integrated operator over the tail of the lag distribution. The coexistence of normal diffusion in the structural subspace and superdiffusion in the bulk, within the same process and governed by a single memory kernel, is natural in the Langevin framework [28, 29] but has not previously been identified empirically in financial data.

D Universality class and intrinsic market scales

The exponent triple $(\hat{\beta}_s, \hat{\beta}_b, \alpha)$ defines the dynamical signature of the operator-space framework. Across all liquid assets studied, $\hat{\beta}_s$ is consistent with normal diffusion ($\hat{\beta}_s \approx 1$), $\hat{\beta}_b$ is superdiffusive ($\hat{\beta}_b \approx 1.3 - 1.7$), and α is concentrated in the range $[1.87, 2.45]$, close to the value of 2 associated with interface depinning and record-breaking statistics [18, 17, 30]. The optimal parameters $(\Delta\tau_{\text{opt}}, L_{\text{opt}})$ vary across assets (Table ??), reflecting intrinsic dynamical scales of each market; their variation occurs within a well-defined range, providing evidence for a universal dynamical class in operator-space financial econophysics rather than a fixed-point universality in the strict sense.

To calibrate the position of each asset within this class, we compare the measured α against the AR(1) reference curve described in Supplementary Section ??, which interpolates monotonically between the white-noise ceiling ($\alpha_0 \approx 2.58$, $\phi = 0$) and the GBM floor ($\alpha_{\text{GBM}} \approx 1.87$, $\phi = 1$). All liquid assets fall within this band, spanning $\hat{\varphi}_{\text{eff}} \in [0.1, 0.9]$. Most cluster in the range $\hat{\varphi}_{\text{eff}} \in [0.5, 0.9]$, indicating substantial proximity to the GBM limit; however, two equity assets (ABEV3, BBDC4) yield $\hat{\alpha} \approx 2.45$, placing them near the white-noise end of the reference band and suggesting lower intraday spectral persistence in their return dynamics. Among liquid assets, BIT\$D (Bitcoin futures) yields the lowest $\hat{\alpha} \approx 1.94$ (M5) and ≈ 1.82 (H1), the closest to the GBM floor ($\hat{\varphi}_{\text{eff}} \approx 0.89$), indicating that this market exhibits the least detectable departure from random-walk spectral dynamics at the timescales analysed—consistent with high intraday efficiency in a continuously traded, globally accessible asset.

The procedure for identifying these exponents is analogous to focusing an optical instrument: the exponents are the physical observables, and $(\Delta\tau_{\text{opt}}, L_{\text{opt}})$ are the settings required to resolve them without aberration from finite-window or finite-step artifacts. For liquid assets with continuous price formation, these parameters reflect intrinsic dynamical scales of $K(\tau)$, set by the autocorrelation structure of the operator rather than by the observation frequency — a property confirmed by their stability across M5 and H1, which differ by a factor of 12 in physical time. For assets with intermittent or structurally low liquidity, the same criteria yield effective parameters that absorb microstructural contributions. In these cases $\hat{\beta}_{\text{struct}}$ remains well-defined, but its interpretation requires accounting for the non-uniform sampling of the underlying continuous process. Notably, the tail exponent $\hat{\alpha}$ estimated for low-liquidity assets tends to exceed that of liquid ones — with values reaching $\hat{\alpha} \approx 2.5-3.7$ — indicating that structural transitions in the spectral gap become rarer and less correlated, moving the return-time statistics toward the white-noise limit. The liquid and low-liquidity regimes are analysed separately in Section ??.

The automated criterion recovers the visually identified plateau region to within one grid step for all assets in the liquid sample (Table ??). Residual discrepancies of $\sim 1-2$ steps occur for assets where the M5 and H1 curves are systematically offset but individually flat, reflecting the intrinsic uncertainty in plateau detection when inter-scale coherence is limited by microstructural differences between time-frames. In such cases the reported $\Delta\tau_{\text{opt}}$ should be understood as the lower bound of a plateau interval rather than a sharp optimum, and the sensitivity of the downstream exponents to this ambiguity is assessed in Supplementary Section ??.

E Scope and extensions

The operator-space framework introduced here is not specific to financial markets. Any multivariate time series admitting a Hankel embedding can be analysed within the same formalism, and the cross-

scale calibration procedure applies without modification. The extension to other complex systems like turbulence, neural population dynamics, ecological and climate time series, where anomalous diffusion and non-trivial spectral structure coexist [28, 29], represents a natural direction for future investigation.

VIII Conclusion

We have introduced a framework in which the temporal covariance operator $K(\tau)$, constructed via Hankel embedding of the return series, is the fundamental dynamical object of financial markets. The observed price return is the spectral projection of $K(\tau)$, decomposable into a structural signal subspace and a Marchenko-Pastur bulk via Eq. (1). The dynamics of $K(\tau)$ are governed by a generalised Langevin equation with a quasi-block-diagonal memory kernel, from which three universal exponents emerge. $\hat{\beta}_s \approx 1.0$ characterises near-Brownian diffusion of the structural modes, with the small excess over unity attributable to weak inter-block coupling and Poisson shocks. $\hat{\beta}_b \approx 1.3\text{--}1.7$ characterises anomalous superdiffusion of the bulk modes, driven by collective Dyson-type eigenvalue mixing and a divergent memory integral. $\alpha \approx 2$ characterises the universal power-law distribution of inter-regime waiting times, consistent with first-passage theory at a regular critical point and with the boundary of the CTRW transition [29, 28].

Together with the intrinsic scales L_{opt} and $\Delta\tau_{\text{opt}}$, these exponents are consistent across 14 assets, five asset classes, and two intraday timescales. Building on the anomalous diffusion programme of Refs. [14, 27] and extending the static RMT framework of Refs. [7, 8, 2], the present work establishes a direct bridge between the physics of stochastic operators and the phenomenology of financial markets, and places financial phase transitions in a well-defined universality class. The rigorous derivation of $\alpha = 2$ from the Dyson-Langevin dynamics of the spectral edge process, and the extension of the framework to other complex multivariate systems, remain as well-posed problems for future work.

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